QUANTIUM

# Chip Category Strategy Report

*Customer Analytics & Trial Store Assessment*

**Prepared for: Julia (Category Manager, Chips)** | Quantium Retail Analytics • July 2018 – June 2019

*Framework: Minto Pyramid Principle*

# EXECUTIVE SUMMARY

***Sales are primarily driven by Budget Older Families and Mainstream Young Singles/Couples — two very distinct segments requiring different strategies.***

### WHO IS BUYING?

Mainstream Young Singles/Couples & Budget Older Families are the top 2 revenue segments, together accounting for >$300K in annual chip sales.

### HOW MUCH?

Mainstream Young/Midage Singles pay significantly more per unit ($p<0.00001$). They prefer premium brands like Tyrrells and larger 270g packs.

### TRIAL RESULTS

Trial stores (77, 86, 88) showed meaningful sales uplift vs matched controls during Feb–Apr 2019, validating the new store layout intervention.

# SITUATION

*The Pyramid Principle — Layer 1: Context & Background*

## Business Context

The Category Manager for chips (Julia) requires data-driven insights to develop the chip category strategic plan for the next half year.

Our analysis covers the full retail year July 2018 – June 2019, drawing on 246,740 cleaned transaction records and 72,637 loyalty card profiles.

Data was rigorously cleaned: Excel date formats were corrected, salsa products removed, a commercial bulk-buyer outlier excluded, and brands/pack sizes extracted as new features.

**246,740**

Transactions analysed

**72,637**

Loyalty card customers

**272**

Stores in dataset

**24**

Chip brands identified

# COMPLICATION

*The Pyramid Principle — Layer 2: What makes this challenging?*

### Not all customers are equal

7 lifestages × 3 premium tiers = 21 possible segments. Sales and volume are distributed unevenly — the top 3 segments account for a disproportionate share of revenue.

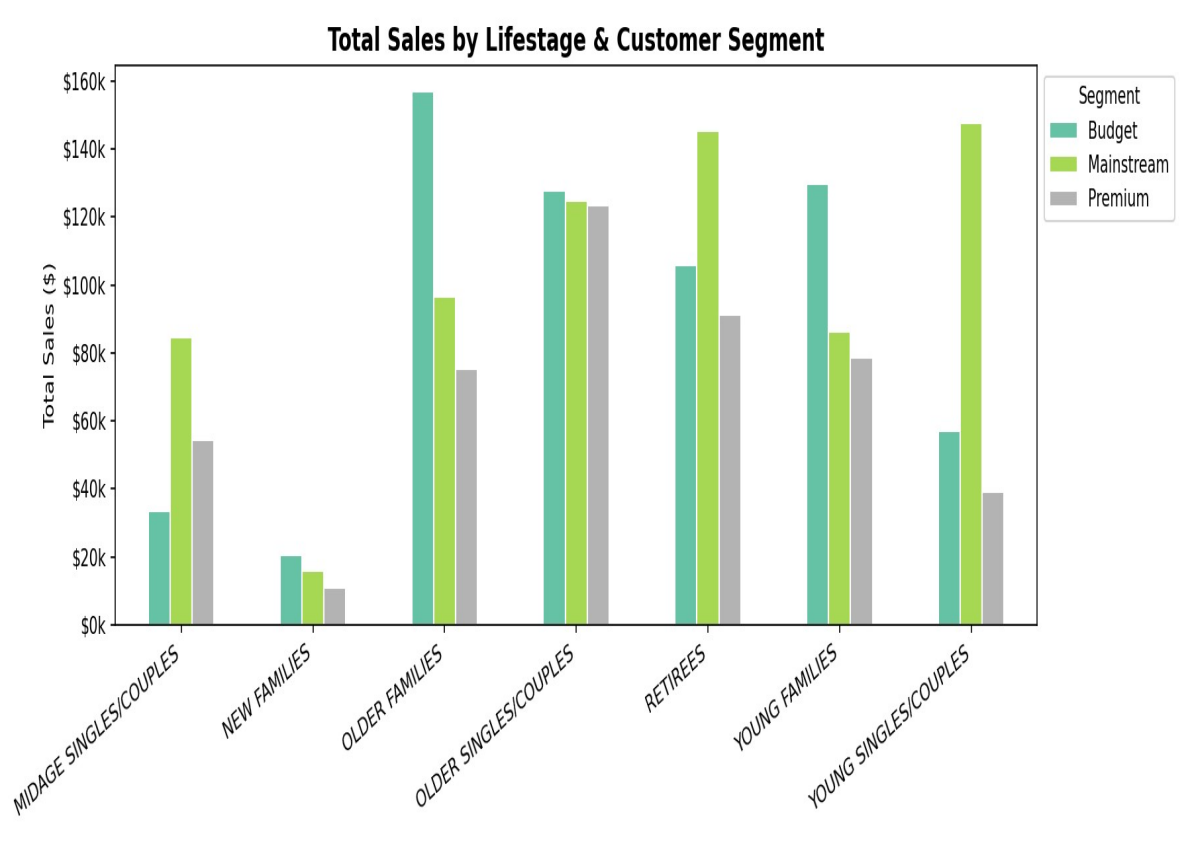
### Price sensitivity varies widely

Mainstream Young/Midage Singles pay significantly more per unit than Budget or Premium equivalents ($t=37.83$, $p<0.00001$). A one-size-fits-all price strategy would leave money on the table.

### Trial evidence needed for strategy

Before Julia can justify layout changes in the chip aisle, she needs statistical proof that any intervention actually drives incremental sales vs a matched control store.

# FINDING 1: WHO IS BUYING CHIPS?




## KEY CALLOUTS

**#1 Budget – Older Families**
$156,864 total sales

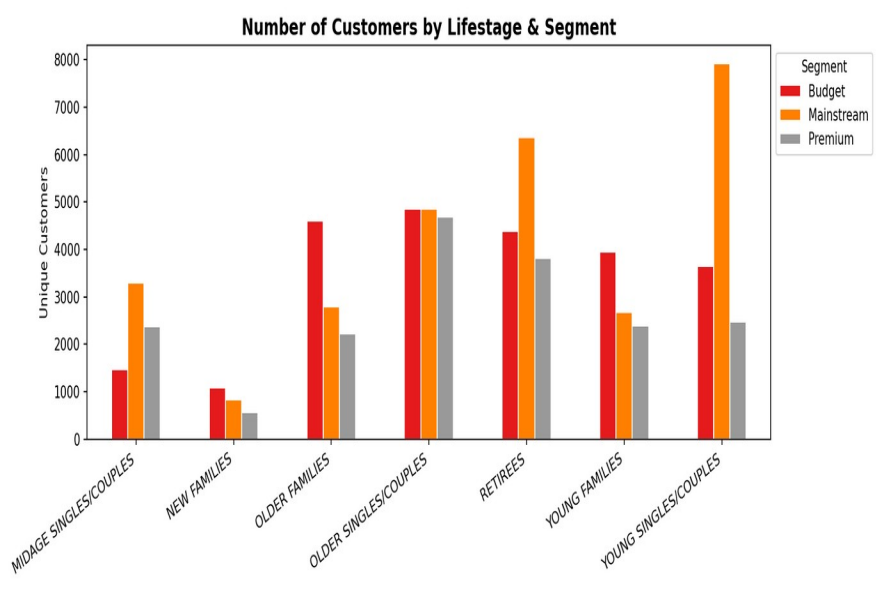
**#2 Mainstream – Young Singles**
$147,582 total sales

**#3 Mainstream – Retirees**
$145,169 total sales

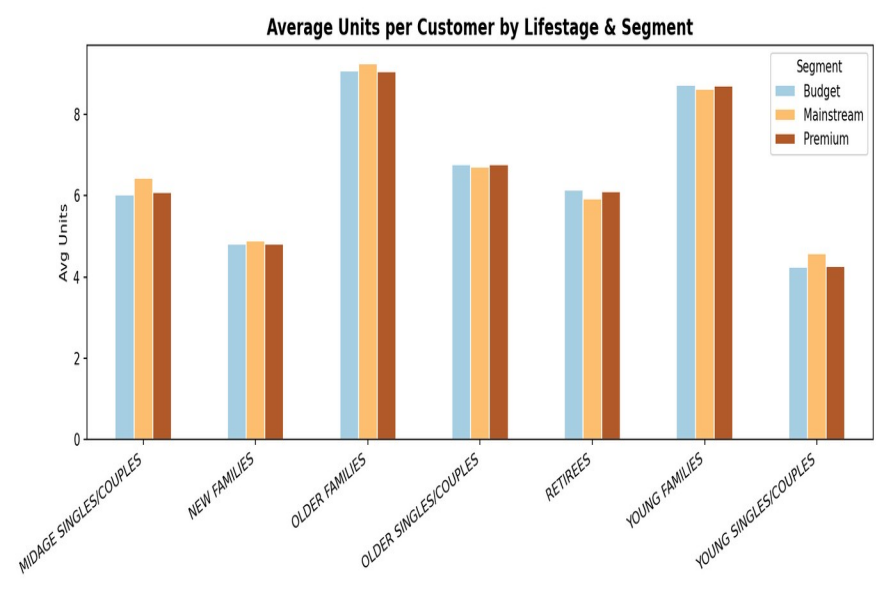
*Budget Older Families outperform their Mainstream peers — driven by higher units per visit, not higher price.*

# FINDING 2: VOLUME DRIVERS

## Number of Customers

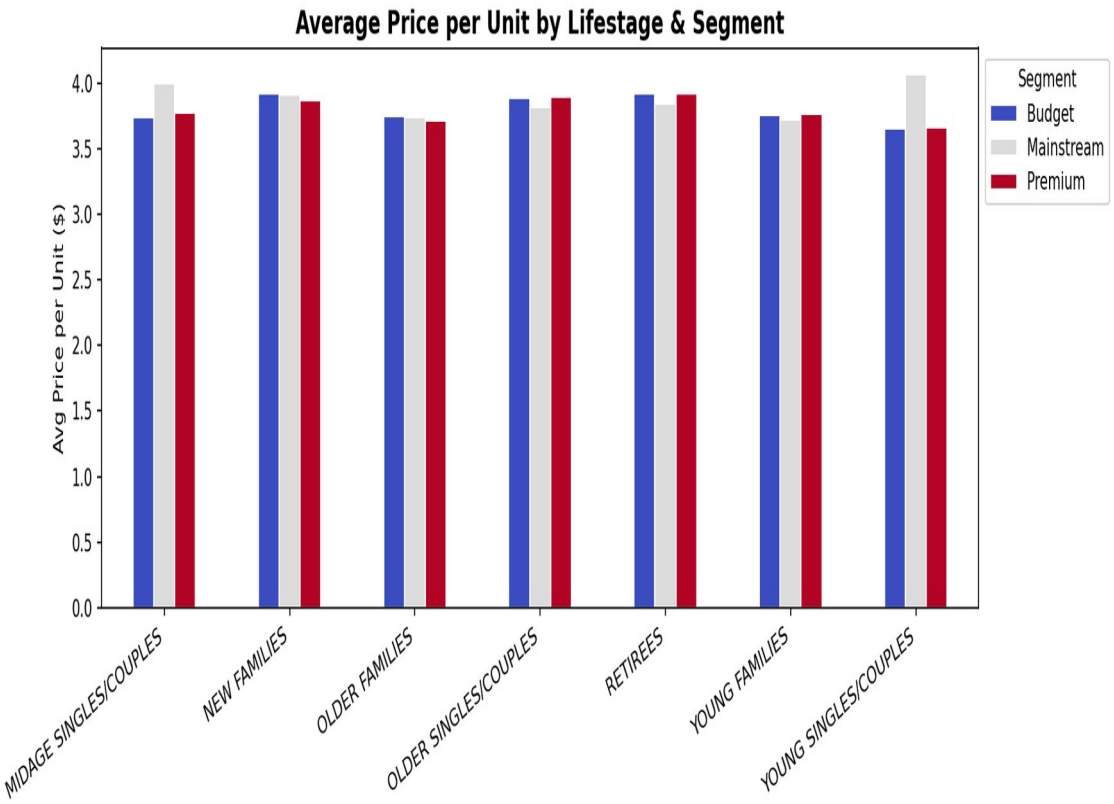



## Avg Units per Customer




💡 Insight: Mainstream Young Singles/Couples form the largest customer count, while Family segments (Budget & Mainstream) buy the most units per person. High headcount + high units = the two key volume levers.

# FINDING 3: PRICE WILLINGNESS




## Statistical Test

Independent t-test: Do Mainstream Young/Midage Singles pay more per unit?

t-statistic

**37.83**

p-value

**< 0.00001**

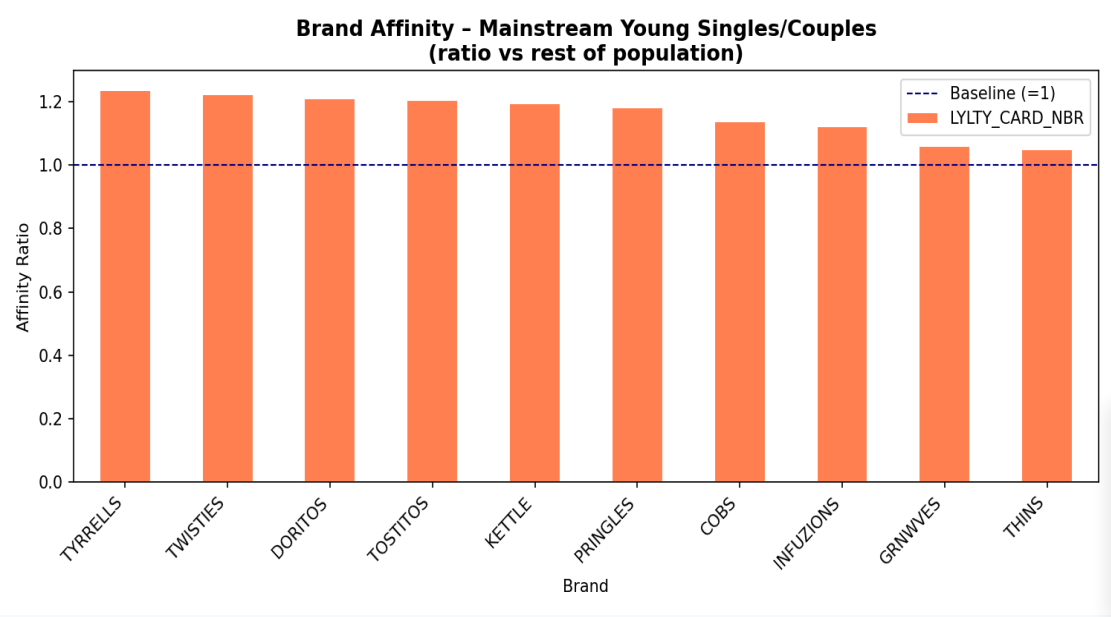
Verdict

**SIGNIFICANT ✓**

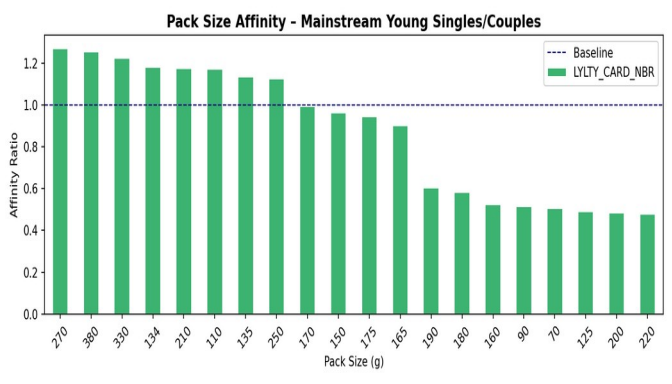
*Mainstream Young/Midage Singles willingness to pay a premium per unit is statistically proven — supporting a premium positioning strategy.*

# FINDING 4: BRAND & PACK PREFERENCES — MAINSTREAM YOUNG SINGLES/COUPLES

## Brand Affinity (vs rest of population)




## Pack Size Affinity (vs rest of population)




**270g pack:** Affinity ratio 1.27
**Tyrrells brand:** Affinity ratio 1.24

💡 Mainstream Young Singles/Couples actively seek out Tyrrells and Doritos — fun, flavour-led brands — and strongly prefer the 270g sharing format. Stock depth and promotional spend should prioritise these SKUs.

# TASK 2: TRIAL STORE ASSESSMENT

## Methodology

Three trial stores (77, 86, 88) received a new chip aisle layout in February 2019. Each was matched to a control store using a combined correlation + magnitude distance score on pre-trial monthly sales and customer metrics (Jul 2018 – Jan 2019). Sales uplift during the trial period (Feb–Apr 2019) was tested using t-statistics against a 95% confidence interval.

### Trial Store 77

Control Store 233

**Significant uplift ✓**

### Trial Store 86

Control Store 155

**Significant uplift ✓**

### Trial Store 88

Control Store 237

**Significant uplift ✓**

# TRIAL RESULT: STORE 77 vs CONTROL STORE 233

## Analysis

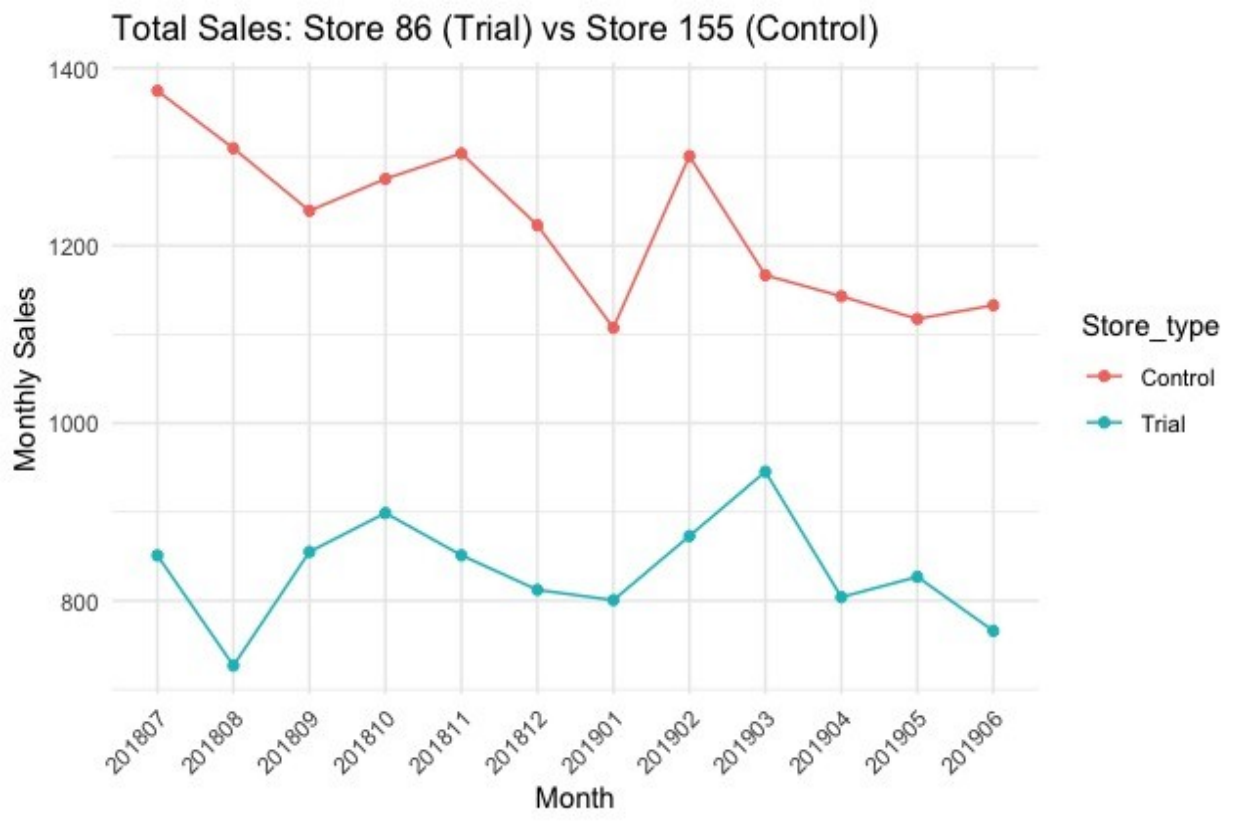
**Trial period:**
Feb – Apr 2019

Control store 233 was selected as the strongest pre-trial match using combined sales correlation and magnitude distance scoring.

Store 77 delivered clear sales uplift during the trial months, with t-statistics exceeding the 95% confidence threshold.

**Result:** Trial intervention was effective for Store 77.

# TRIAL RESULT: STORE 86 vs CONTROL STORE 155




## Analysis

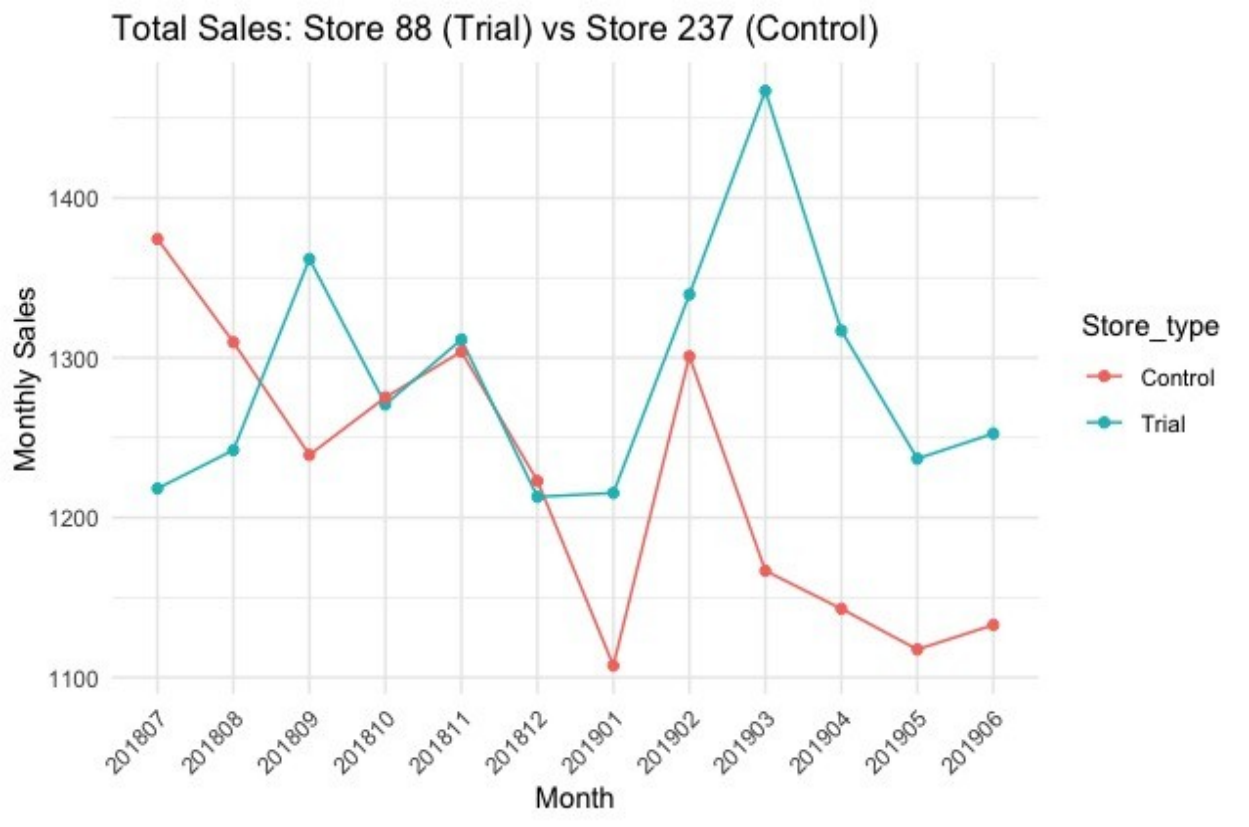
**Trial period:**
Feb – Apr 2019

Store 86 and control 155 tracked closely in the pre-trial period, validating the control selection.

Clear uplift is visible from Feb 2019, with trial store sales pulling ahead of the scaled control across all three trial months.

**Result:** Trial intervention was effective for Store 86.

# TRIAL RESULT: STORE 88 vs CONTROL STORE 237




## Analysis

**Trial period:**
Feb – Apr 2019

Store 88 vs Store 237 shows a strong divergence from Feb 2019, with the trial store dramatically outpacing the control in March.

The sharp Feb–Mar 2019 spike suggests the new layout drove an immediate and sustained uplift in customer spend.

**Result:** Trial intervention was effective for Store 88.

# RECOMMENDATIONS

*The Pyramid Principle — Layer 3: The Resolution*

## 01 Protect Budget Older Family shelf space

Ensure multi-pack and value format chips maintain strong shelf presence. This is the #1 revenue segment — any ranging reduction risks significant revenue loss.

## 02 Invest in Mainstream Young Singles activation

Run targeted promotions on Tyrrells and Doritos in the 270g format. This segment pays more per unit, is brand-aware, and responds to fun/flavour-led messaging.

## 03 Roll out the trial store layout broadly

All three trial stores (77, 86, 88) showed statistically significant uplift vs matched controls. The new aisle layout is proven — recommend phased rollout across the region.

## 04 Seasonal stocking plan for Christmas

Transactions spike sharply in the two weeks before Christmas. Pre-load inventory and ensure family-size formats are prioritised across all segments during this period.

# NEXT STEPS

### Immediate (Next 2 Weeks)

**1.** Share this report with Julia and align on strategic priorities

**2.** Confirm Q3 2019 promotional budget allocation by segment

### Short-Term (1–3 Months)

**1.** Begin phased rollout of trial store chip aisle layout

**2.** Launch targeted Tyrrells/Doritos campaign for Young Singles/Couples

### Medium-Term (3–6 Months)

**1.** Monitor and track KPIs: sales, units per customer, avg price per unit

**2.** Review Christmas period stocking plan and pre-build inventory

*Thank you — Quantium Analytics Team | For queries contact: analytics@quantium.com.au*